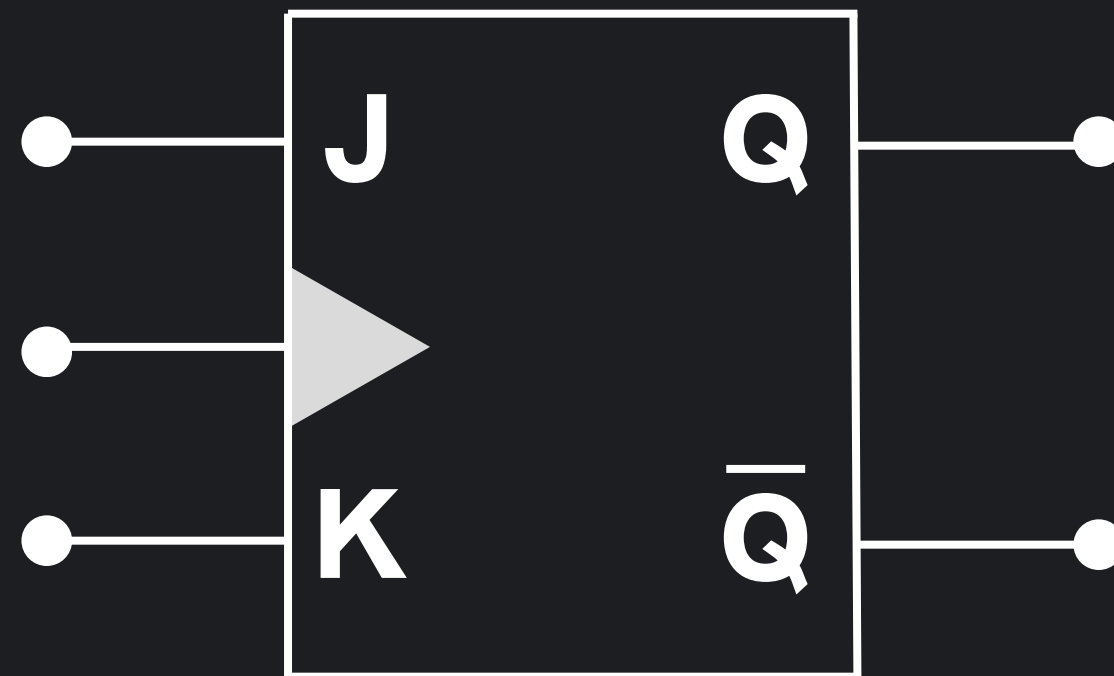


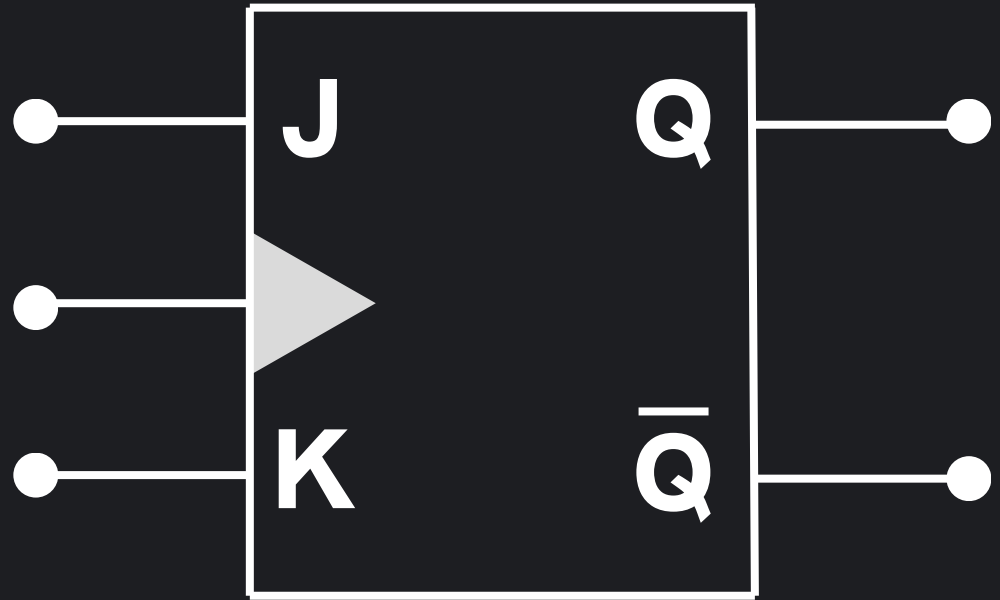
JK-FLIP FLOP



What is a JK-FLIP FLOP?

A JK Flip-Flop is a basic sequential logic gate that produces an output based on its inputs and its previous state. The output is in a Toggle state when both inputs (J and K) are HIGH (1), while it maintains its previous state when both inputs are LOW (0).

Truth Table



J	K	CK	Q	made of operation
x	x	↑	Q_0	NO Change
0	0	↓	Q_0	NO Change
0	1	↓	0	Resets
1	0	↓	1	Sets
1	1	↓	Q_0	Change



Function of the JK-FLIP FLOP

A JK Flip-Flop performs a logical storage and Toggle operation on its inputs. The main function of a JK Flip-Flop is to serve as a basic memory element and produce an output signal according to this rule:

Output = Toggle when both inputs J and K = 1

Output = No Change when both inputs J and K = 0

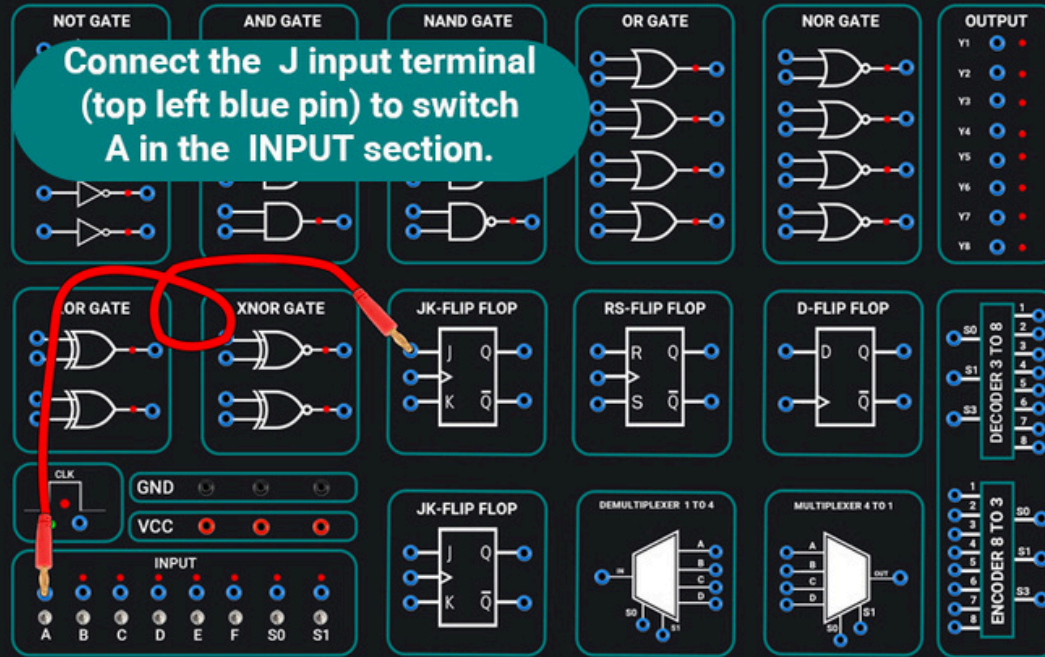
Output = 1 (Set) when J = 1 and K = 0

Output = 0 (Reset) when J = 0 and K = 1

1

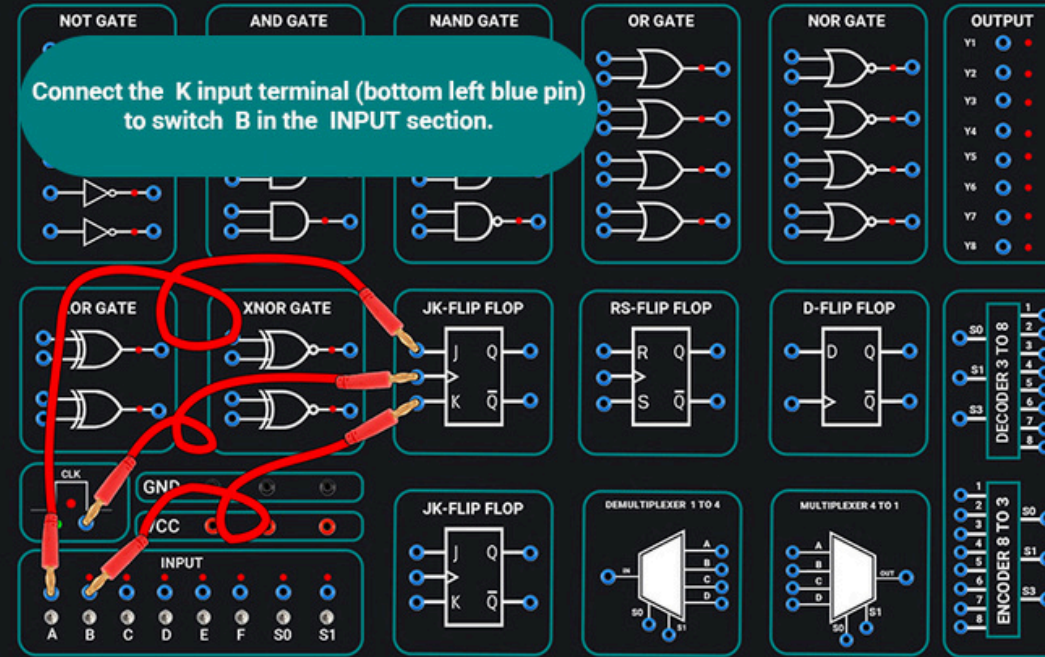
SMART DIGITAL CIRCUITS BOARD

Connect the J input terminal (top left blue pin) to switch A in the INPUT section.



SMART DIGITAL CIRCUITS BOARD

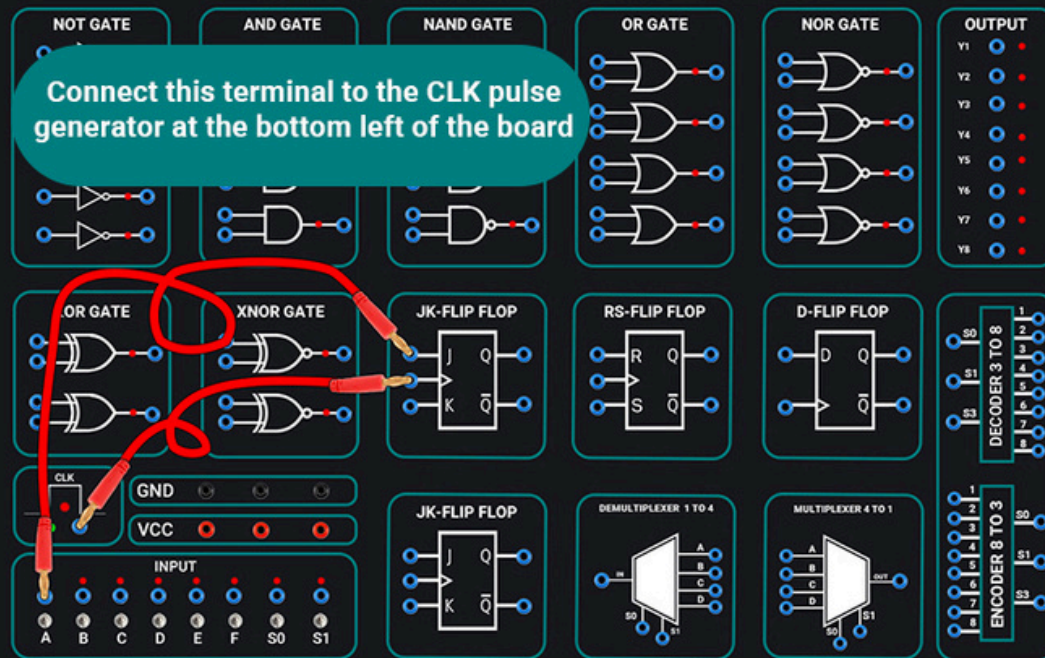
Connect the K input terminal (bottom left blue pin) to switch B in the INPUT section.



3

SMART DIGITAL CIRCUITS BOARD

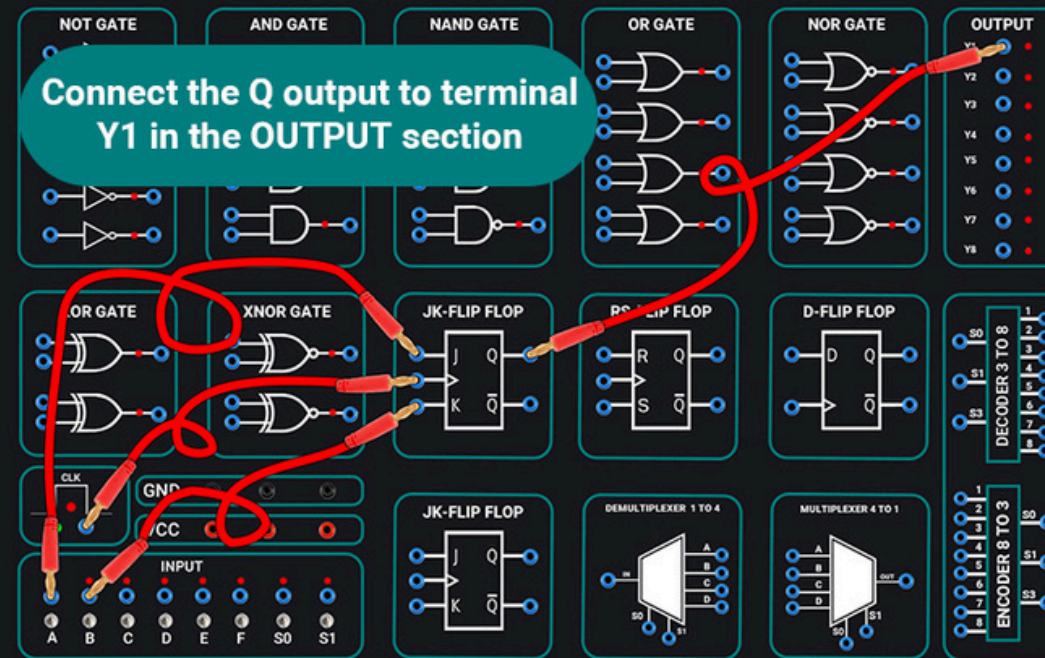
Connect this terminal to the CLK pulse generator at the bottom left of the board



2

SMART DIGITAL CIRCUITS BOARD

Connect the Q output to terminal Y1 in the OUTPUT section



4

SMART DIGITAL CIRCUITS BOARD

Connect the complementary output $\bar{Q}$ to terminal Y2.

5

